

Research

A Qualitative Approach to Understanding the Effects of a Caring Relationship Between the Sonographer and Patient

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ABSTRACT

Background: The concept of caring is rapidly becoming an area of interest in healthcare research, especially in medical imaging. While patients, academic institutions, hospitals and policy makers have an expectation for healthcare professionals to be caring, the effects of such a relationship between patient and professional is equally important to understand. Within sonography, a discipline of radiography, there is limited literature on this concept.

Purpose: The purpose of the main study was to explore and describe the sonographers' experiences of being caring professionals within private practices in the province of Gauteng, South Africa. This paper focuses on one essential theme that emerged from the study: the effects of a caring relationship between sonographer and patient. Guidelines to enhance caring relationships are also provided.

Materials and methods: A qualitative, exploratory, descriptive, contextual, phenomenology research design was utilised. Four focus group interviews (n = 14) were conducted with qualified sonographers working in the private sector, in Gauteng. Data collection took place until data saturation was reached.

Results: Sonographers described caring for their patients as having both positive and negative effects on themselves. Sonographers explained having developed protective mechanisms to help them when caring for tenacious patients. There was also a sense of emotional and psychological strain identified. Despite this, sonographers displayed a sense of professional pride for their careers.

Conclusion: Guidelines were developed to alleviate the emotional and psychological strains sonographers are burdened with. These

guidelines included; the facilitation of a supportive and healing environment through reflective practice, availability of a psychologist, and having a tea garden at work.

RÉSUMÉ

Contexte : La notion de soins attentionnés devient rapidement un champ d'intérêt en recherche sur la santé, et plus particulièrement en imagerie médicale. Alors que les patients, les institutions d'enseignement, les hôpitaux et les décideurs politiques s'attendent de plus en plus à ce que les professionnels de la santé se montrent attentionnés, Il devient tout aussi important de bien comprendre les effets d'une telle relation entre les patients et les professionnels. En échographie, une branche de l'imagerie médicale, peu d'articles ont été écrits sur cette question.

But : L'étude principale visait à explorer et décrire les expériences des échographistes à titre de professionnels attentionnés dans les cliniques privées de la province de Gauteng, en Afrique du Sud. Cet article met l'accent sur un thème essentiel qui est ressorti de l'étude: les effets d'une relation attentionnée entre l'échographiste et le patient. Des lignes directrices pour améliorer ces relations sont également fournies.

Matériel et méthodologie : Une étude phénoménologique contextuelle qualitative, exploratoire et descriptive a été utilisée. Quatre groupes de discussions (n = 14) ont été tenus avec des échographistes qualifiés œuvrant dans le secteur privé, dans la province de Gauteng. La collecte de données a eu lieu jusqu'à ce que le point de saturation des données soit atteint.

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Ethical Approval: Informed consent was obtained from all the participants. Ethical clearance to conduct the study was obtained from the Research Ethics Committee of the Faculty of Health Sciences at the University of Johannesburg (REC-01-34-2018).

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Résultats : Les échographistes ont témoigné que le fait d'offrir des soins attentionnés aux patients avait des effets positifs et négatifs. Les échographistes ont expliqué avoir développé des mécanismes de protection pour les aider à soigner des patients tenaces. Ils ont également identifié des tensions émotionnelles et psychologiques. Malgré cela, les échographistes affichent une fierté professionnelle envers leur carrière.

Keywords: Caring; burnout; compassion fatigue; professional pride; qualitative; sonographer

Introduction

Being a healthcare professional is considered to be a “calling”.¹ Paulson² and Bolderston et al.³ define a caring professional as a person who displays a humanistic approach of compassion, empathy and kindness towards patients, while optimally operating diagnostic equipment. Care theorist and educational philosopher, Nel Nodding, argues that caring is the core of being human, it is more than an attitude of a single moral agent and not only the behaviour of a “good person”.^{4,5}

The continuous interaction with patients can be an immensely rewarding experience and it is what keeps many in this field.¹ Professional pride is a strong driving force behind healthcare professionals. From a patient's perspective, having healthcare professionals who care about competence in their role and have pride in their work, is comforting.⁶ However, when caring is absent, dissatisfaction can occur and the patient feels like an object.⁷ The demand for delivering high-quality healthcare services has increased tremendously throughout recent years and consequently resulted in increased stress among healthcare professionals.⁸ Moreover, the objectification of patients is on the rise and therefore removes the humanity aspect of the patient.⁹ According to Reeves and Decker⁹ and Naidoo et al.,¹⁰ objectification is used as a coping mechanism. By objectifying the patient, healthcare professionals avoid getting too attached to the patient.

While the physical and emotional needs of patients are crucially important, the empathy and compassion that are shown by healthcare professionals can be physically and psychologically destructive.^{1,11} Current literature reveals that negative and stressful work environments and an increase in workload are critical factors that lead to compassion fatigue and burnout in healthcare professionals.^{1,12} Compassion fatigue is a diminished desire to have compassion and empathy for patients.^{1,12} It is commonly characterised by anger, irritability and exhaustion.¹¹ Similarly, burnout is a psychological syndrome characterised by emotional exhaustion and depersonalisation, hence feeling emotionally and physically depleted and worn-out.¹³

The concept of caring is becoming an interest in healthcare research, especially in radiography. While patients, academic institutions, hospitals, and policy makers have an expectation for healthcare professionals to be caring, the effects of such a relationship between patient and healthcare professional is

Conclusion : Des lignes directrices ont été produites afin d'atténuer les pressions émotionnelles et psychologiques ressenties par les échographistes. Ces lignes directrices portent notamment sur la facilitation d'un environnement de soutien et de guérison par le recours à la pratique réflexive, la disponibilité d'un psychologue et l'installation d'un jardin de thé sur les lieux de travail.

equally important to understand. Sonographers have a special and unique relationship with their patients. Therefore, it is essential to fully understand the effects on the sonographer.

The study aimed to explore and describe the sonographers' experiences of being a caring professional within private practice in the province of Gauteng, South Africa. This paper focuses on one essential theme that emerged from this study: the effects of a caring relationship between sonographer and patient. Additionally, guidelines to enhance the caring relationship were developed.

Materials and Methods

Ethical clearance to conduct the study was obtained from the Research Ethics Committee of the Faculty of Health Sciences at the University of Johannesburg (REC-01-34-2018). The researcher adhered to the ethical principles of autonomy, beneficence, non-maleficence and justice. Before conducting focus group interviews, informed consent was obtained from the interviewees. One consent form was for participation in the research study and another to audiorecord the interviews. Each participant received a full description of the study by means of an information letter. Anonymity in this study was achieved by not documenting the names of the participants.¹⁴ The participants were reminded that their participation was voluntary and that they were entitled to withdraw at any time without penalty.

A qualitative, exploratory, descriptive, contextual, phenomenology research design was used in this study.¹⁵ The purpose was to understand the effects of a caring relationship between the sonographer and patient. This research design was applied to explore and describe sonographers' experiences of being a caring professional.

The research population consisted of sonographers registered with the Health Professions Council of South Africa (HPCSA) working in private practices in Gauteng. All the participants were female who worked in urban areas. The age of the participants ranged from 27 to 42 years and they were from different cultural backgrounds. The experience of the participants as qualified sonographers ranged from 3 to 16 years. All participants had a Bachelor of Technology degree in diagnostic ultrasound acquired from a higher education institution in South Africa.

Sonographers were invited to participate in this study through purposive snowball sampling to ensure information-rich data. The sample size was dependent on

data saturation. Data saturation, as described by Du-Plooy-Cilliers et al¹⁴ and Holloway and Wheeler,¹⁶ means that no new or additional information originates from the data being collected.

Data Collection

The data collected focussed on sonographers' understanding of caring. The participants were provided with an information letter in order to explain the background of, rationale for and purpose of the research study. Informed consent was obtained for participation and audio recording before conducting the interviews. The focus group interviews were conducted outside the radiology department in a neutral environment that was convenient for the participants. Audiorecordings were used to pose a central question, that allowed the participants to express their feelings within a group. The research question that guided the study was: "*Tell me about being a caring professional in sonography*". The focus group interview began with a broad, open-ended question, and depending on responses received, further probing questions were asked to obtain meaningful data. Extensive field notes were taken on additional information such as body language and group dynamics.¹⁷ The utilisation of probing questions and paraphrasing added value to these interviews until data saturation was reached. The researcher used probing questions to gain more insight into sonographers' experiences of being a caring professional in sonography and to preserve the line of investigation.^{18,19} Probing questions were also used to encourage the interviewees to elaborate on their answers.^{18,19}

Four focus group interviews were conducted and the groups ranged between 3 and 5 participants. The groups varied in size due to the difficulty to recruit more sonographers to attend interviews at the same time and place. A total of 14 participants were interviewed and each interview lasted between 20 and 60 min. Data collection took place until no new information was obtained from the focus group interviews and the researcher saw similar results appearing after each interview. After the third interview data saturation was reached and no new information was shared. The researcher then conducted a fourth interview as a control to confirm this finding. Data were collected between June and July 2018. After data collection, full transcription followed that documented non-verbal communication such as mannerisms, tone and emotional content.²⁰

Analysis

The researcher analysed the data in the form of coding for themes and categories. Thereafter, memos were made about the context of the variations in the phenomenon under study. The selected themes were verified through reflection on the data categories, recording support data for categories and identifying propositions.¹⁸ Coding and categories were initiated as soon as data collection began and were used to organise data collection in interviews. Coding was re-evaluated by the researcher's supervisors until consensus was reached.

The researcher developed categories through manual analysis by reviewing the recorded information.

Data were transcribed and then analysed by means of thematic analysis by the researcher. This is in keeping with a qualitative research methodology. Thematic analysis is a method of identifying, analysing and reporting themes. This method reduces the amount of data in a more flexible way.²¹ The process of analysis described by Holloway and Wheeler¹⁶ was used, namely transcribing interviews and sorting field notes, organising, ordering and sorting the data and repeatedly listening to and reading or viewing the material collected. The audiotapes of the focus group interviews were transcribed and the researcher analysed the transcribed data alongside the documented field notes.^{18,19} Triangulation through descriptive field notes, reflective field notes and interview data was used to enhance credibility.

Trustworthiness

Decisions on trustworthiness by the researcher were made through developing dependability, credibility, transferability and confirmability.¹⁶ Credibility was obtained by demonstrating that the subject was accurately described and identified.^{22,23} Reflexivity, peer debriefing and triangulation were used to ensure credibility. Transferability in this research study was achieved by providing a comprehensive description of the research setting, data population and methodology. The researcher kept an audit trail and had described the methodology in detail. The aim was to help readers follow the path and to demonstrate how the conclusions were reached. This method ensured dependability. Confirmability in this study was accomplished by confirmability audit procedures. The confirmability audit procedure was done by writing extensive field notes and audiorecording the interviews. The descriptive field notes, reflective field notes and audio-recordings will be kept for two years.

Results

The major theme that emerged from the data was "the effects of a caring relationship between sonographer and patient". Three significant categories made up this theme namely: professional pride, the protective mechanisms of the sonographer, and the emotional/psychological strain of the sonographer. See [Table 1](#) below for a summary of the relationship between the theme and category followed by thorough description.

Theme: the Effects of a Caring Relationship Between Sonographer and Patient

This theme focuses on the participants' feelings of how caring for their patients impacted them. It was evident throughout the interviews that there was a unique relationship between the sonographer and patient. Sonographers were still keen to be caring professionals despite experiencing both positive and negative effects of caring.

Table 1
Themes and Categories Developed from Focus Group Interviews

Theme	Categories
The effects of a caring relationship between sonographer and patient.	1. Professional pride. 2. The protective mechanisms of the sonographer. 3. The emotional/psychological strain of the sonographer.

The participants explained that they enjoyed being able to serve people and they displayed a sense of professional pride during these focus group interviews. Despite the complex situations, the sonographers surveyed were still able to commit to their profession. The effects that caring had on the participants are explained in the categories below:

Category 1: Professional Pride

The participants explained that they enjoy being caring professionals due to their fulfilling careers. Some participants said that they enjoyed giving answers to patients and that they felt happy to make a difference in patients' lives. Despite some participants being introverts, a sense of professional pride was still apparent during these interviews. The participants' quotes below support these findings:

FG 3, P 1: "I think even though I am an introvert, I am still caring. I am very emotional. So I think I do enjoy it. It is fulfilling to be able to serve ..."

FG 3, P 3: "I like giving answers to patients. I believe everybody there wants to know what is going on."

FG 1, P 4: "It made me happy that I could be part of her life, even if it was just for a short while. It made me sad to think what she left behind or who she left behind."

Pride is a powerful force that empowers a high quality of professional work.²⁴ Pride links to having a professional identity.⁹ Literature suggests that the teaching of medical professionalism assists healthcare professionals in developing their own professional identity.²⁵ Sociologist Robert Merton stated in a 1957 study of undergraduate medical education that it is the purpose of medical schools to shape beginners into actual practitioners.^{25,26} This will offer them the best skills and knowledge. A healthcare professional will then ultimately create a professional identity so that they can think, feel and act like a physician.^{25,26} The participants in the study resonated with the findings of current literature because they showed distinctive aspects of professional pride in their profession as sonographers.

Category 2: The Protective Mechanisms of the Sonographer

The participants explained that they had a sincere desire to be compassionate and empathetic. However, they create barriers when encountering difficult situations that may be emotionally depleting. The participants also admitted being watchful about getting too attached with patients for the reason that it may leave them unable to deliver the required

service. The protective mechanisms of the participants are explained in the statements below:

FG 3, P 1: "I try to be friendly always and to be compassionate and empathetic, but as soon as I feel like this is actually going to drain me too much, then I will put a barrier to it or a personal barrier."

FG 3, P 3: "You need to have empathy with your patients, but not over sympathise that you go home at night wondering how your patient is doing. You need to be able to keep that balance and keep your emotions out the room, but to have empathy with your patient's situation."

FG 2, P 2: "... I think there is a thin line between getting too involved where it compromises the service that you deliver and having professional empathy and when I think about most sonographers that I know ..."

Medical careers are notorious for their high psychiatric morbidity and psychological distress.²⁷ Human beings are taught to distance themselves from undesirable conditions. This action is known as "distancing". Similar to the participants in this study, literature suggests that healthcare professionals use distancing as a coping strategy to deal with difficult situations as part of their professional role.⁹ These actions result in meeting the high demand of efficiency, safety and effective needs of their working environment.^{9,28}

However, distancing can easily result in diminished empathy and limited compassion. A decline in empathy has a negative effect on the care of patients and patient satisfaction.²⁷ Consequently, distancing is one of the characteristics labelled as burnout and compassion fatigue.²⁸ Literature proposes that diagnostic radiographers are less emotionally involved with their patients and are therefore using that as a coping strategy and to minimise job stress.⁹ Nevertheless, it is still important to remember that sonographers spend a fair amount of time with their patients during ultrasound procedures. The participants in this study related to the findings in literature through evidence of distancing.

Category 3: The Emotional/Psychological Strain of the Sonographer

Stories of emotional strain were shared throughout the interviews. The participants acknowledged that they would get too involved with patients and shared their apprehension. Some participants desired a short intermission in between patients. The participants' stories are shared in the verbatim quotes below:

FG 2, P 1: "Well, I think if you get too involved with your patient also it gets difficult for a sonographer. Especially if it is sad cases, because you see so many sad cases, you can't take all of that on yourself. So that can be also a bit of a difficult situation to care too much."

FG 3, P 1: "So it is a very fine line and I think it is probably one of my biggest struggles personally in the career to keep that balance between caring and doing what I need to do ..."

FG 4, P 1: "I think I would like to take more of a timeout really if I am feeling like I am very upset. I think that will help me to come back as a better person. Like even taking a walk if you are really really upset. Just take a walk and cool down so that when you go into the next patient you are coming in fresh ..."

Burnout and emotional exhaustion are regularly detected in the healthcare setting.²⁹ Burnout is defined as a psychological syndrome and is characterised by low job performance, cynicism and emotional exhaustion.³⁰ Studies revealed that patients' needs and workload are principal cause of reduced emotional well-being among healthcare professionals.^{8,29,30} Janjhua and Chandrakanta⁸ conclude that a person's attitude and reaction during complex encounters is determined by their physiological and social composure. These study findings are therefore in line with current literature.

Recommendations

The participants had an overall optimistic outlook and portrayed a sense of professional pride. The researcher felt there was a need to develop recommendations to help facilitate a supportive and healing environment in order to overcome the negative effects of a caring relationship. Recommendations are discussed below:

Reflective Practice

Literature indicates that caring diminishes over time.³¹ The researcher therefore believes that reflection among qualified sonographers and students should be encouraged. Frantz and Rowe³² define reflection as the process of thinking about previous experiences in order to improve current behaviour. They consider it a vital aspect that develops thinking skills. Sonographers should therefore be encouraged to keep reflective journals to document events that influence them emotionally and how they can improve their future reactions towards those events. The reflective journals can then be shared during peer discussions that must be held on a regular basis. The end result of reflective practice should enhance patient experience by having a sonographer who reflects, thinks and cares better for every patient.

Peer Discussion

Sharing knowledge and practical experiences during peer discussion can develop and improve reflective skills and self-

efficacy.^{32,33} Some other benefits of peer discussion include promoting teamwork, leadership and communication. Successful peer discussions have effective advantages such as belonging, involvement and participation.^{34,35} Peer discussion therefore ultimately enhances personal, intellectual and professional development.³⁵

During peer discussions, sonographers can share both difficult and enjoyable encounters. In this way they can propose possible strategies to counteract negative behaviour or challenges and also motivate their peers to be caring sonographers by sharing their heartfelt experiences.

Services of a Psychologist

As indicated in throughout the article, the emotional well-being of healthcare professionals is fundamental. Healthcare professionals work in an environment where they are required to interact with other professionals to ensure the best interest of the patient.²⁹ Sonographers are exposed to numerous aspects that can lead to emotional strain and burnout. Being emotionally exhausted can then have a negative consequence on the patient and patient safety.²⁹

In the data collected, there was no evidence of emotional support for the sonographers. Sonographers did not have access to a psychologist to share traumatising events. The researcher recognised the necessity to set up meetings with a psychologist. These meetings can take place at regular intervals and will encourage all staff members to utilise the facility. This will allow sonographers to receive professional help in time of need. Not only will this implementation be beneficial for the sonographer, but literature suggests that a healthcare professional with good psychological wellbeing will react constructively during the inevitable stressful situations within a healthcare environment.⁸ They will also be able to still care for patients and this caring will not be influenced by their mood that day, or the demanding situations of the department.

Tea Garden

There is growing evidence to suggest that exposure to natural environments can be associated with mental health benefits.³⁶ Exposure to nature has been linked to lower levels of stress and a reduction in anxiety and depression.³⁶ Giving sonographers access to a tea garden will give them the time and ability to relax and meditate. This will maintain psychological well-being by enabling sonographers to de-stress. The patient will automatically benefit from this technique, because sonographers will return to the department less overwhelmed and stressed, thus able to provide sufficient care to the patient.

Limitations of the Study and its Implications

The only limitation that might pose a threat is that this qualitative, descriptive phenomenological study was limited to sonographers who worked in private practices in Gauteng. Hence, the results obtained from the sample of this study cannot be generalised to a larger population. However, that

results can be applicable to other sonographers with a similar context. The researchers experienced no other challenges.

Conclusion

This paper explored and described the effects of a caring relationship between sonographer and patient. It is evident throughout literature that healthcare professionals are susceptible to compassion fatigue and burnout. Despite these emotional and psychological challenges sonographers still exhibited professional pride and satisfaction. Guidelines and recommendations were developed that might not only improve caring among sonographers, but also enhance the psychological well-being of healthcare professionals. Implementation of the guidelines provided is strongly recommended. It is the responsibility of sonographers, sonography departments and universities to implement, enhance and sustain caring within healthcare.

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